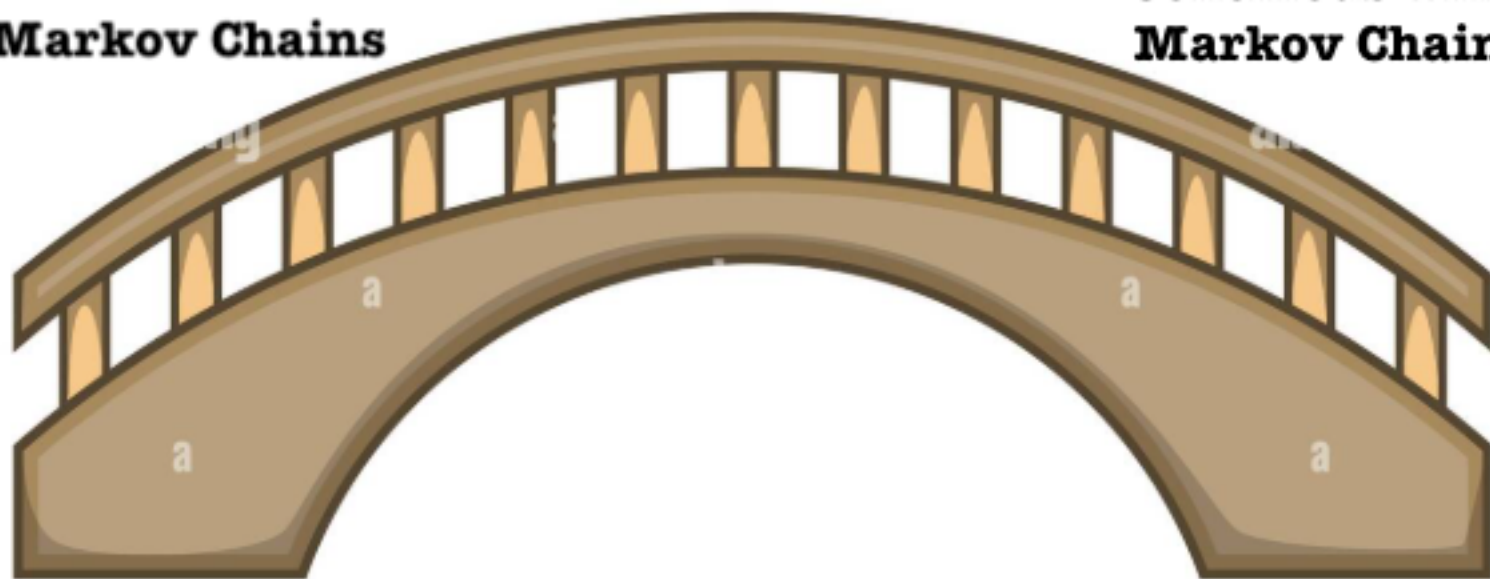


Week 38: Poisson Processes

Poisson Processes

**Discrete-Time
Markov Chains**

**Continuous-Time
Markov Chains**



Definition 1.1. (*Homogeneous Poisson process*) A Poisson process with range $\lambda > 0$ is an integer-valued stochastic process $\{X(t) : t \geq 0\}$ for which:

1. For any time points $0 = t_0 < t_1 < \dots < t_n$, the increments

$$X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$$

are independent.

2. For any $s \geq 0$ and $t > 0$, $X(s + t) - X(s) \sim \text{Poisson}(\lambda t)$.
3. $X(0) = 0$.

Remarks

- $X(t) = X(t) - X(0) \sim \text{Poisson}(\lambda t)$.
- $\text{Var}[X(t)] = \mathbb{E}[X(t)] = \lambda t$.

Problem

Arrival of customers follow a homogeneous Poisson process with rate $\lambda = 4$ customers per hour. The store opens at 09:00. Calculate the probability that exactly one customer has arrived by 09:30 and exactly five customers have arrived by 11:30.

Definition 1.2. (*Inhomogeneous Poisson process*) An inhomogeneous Poisson process with rate $\lambda(t) \geq 0$ with $t \geq 0$ is an integer-valued stochastic process $\{X(t) : t \geq 0\}$ for which

1. Independent increments.
2. For $s \geq 0$ and $t > 0$, $X(s + t) - X(s) \sim \text{Poisson}\left(\int_s^{s+t} \lambda(x) dx\right)$.
3. $X(0) = 0$.

Remarks:

- $\lambda(t) = \lambda_0 \implies$ Homogeneous Poisson process.
- Intuition is area under the curve (x-axis is time, y-axis is rate).

Problem

Arrival of customers follow a homogeneous Poisson process with rate $\lambda = 4$ customers per hour. The store opens at 09:00. Calculate the probability that exactly one customer has arrived by 09:30 and exactly five customers have arrived by 11:30.

Arrival of customers follow an inhomogeneous Poisson process with rate $\lambda(t) = t$. The store opens at 09:00. Calculate the probability of no arrivals by 10:00.

Definition 1.3. (*Waiting time*) The waiting time W_n is the time of the occurrence of the n -th event. We define $W_0 = 0$.

Definition 1.4. (*Sojourn time*) The differences $S_n = W_{n+1} - W_n$ are called sojourn times (or interarrival times).

The intuition is that S_n is the time spent in state n .

Definition 1.5. (*Exponential distribution*) A random variable Y has an exponential distribution with rate $\lambda > 0$ if

$$f(y) = \lambda e^{-\lambda y}, \quad y > 0.$$

We write $Y \sim \text{Exp}(\lambda)$. Have $\mathbb{E}[Y] = 1/\lambda$ and $\text{Var}[Y] = 1/\lambda^2$.

Definition 1.6. (*Gamma distribution*) A random variable Y has a gamma distribution with shape parameter $\alpha > 0$ and rate parameter $\lambda > 0$ if

$$f(y) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y}, \quad y > 0.$$

We write $Y \sim \text{Gamma}(\alpha, \lambda)$. Have $\mathbb{E}[Y] = \alpha/\lambda$ and $\text{Var}[Y] = \alpha/\lambda^2$.

Remark:

- $\text{Gamma}(1, \lambda) = \text{Exp}(\lambda)$.
- Be aware of scale parameterization $\theta = 1/\lambda$ used by some authors.

Theorem 1.7. *For a Poisson process with rate λ , we have*

$$S_0, S_1, \dots, S_{n-1} \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$$

for all $n \geq 1$.

Theorem 1.8. *For a Poisson process with rate λ , we have*

$$W_n \sim \text{Gamma}(n, \lambda)$$

for all $n \geq 1$.

Problem

The occurrences of a rare disease follows a Poisson process with rate $\lambda = 2$ per month. Calculate the following:

- The probability that the first case occurs after 1 month.
- The expected time to the 10-th case occurs.

Problem

Let $\{X(t) : t \geq 0\}$ be a Poisson process with rate $\lambda > 0$. Show that

$$W_1 \mid X(t) = 1 \sim U(0, t).$$

Problem

Arrival of customers follow a homogeneous Poisson process with rate $\lambda = 4$ customers per hour. The store opens at 09:00. Calculate the probability that exactly one customer has arrived by 09:30 and exactly five customers have arrived by 11:30.

Arrival of customers follow an inhomogeneous Poisson process with rate $\lambda(t) = t$. The store opens at 09:00. Calculate the probability of no arrivals by 10:00.

Arrivals follow a Poisson process with rate λ . A store opens at 09:00. Calculate the probability that exactly 5 people have arrived by 10:00, conditional that exactly 10 people will arrive by 11:00.

VM 2026 by Norsk Regnesentral

VG **FOTBALL-VM 2026**



 Program

 Profeten

 VM-døgnet

 Mer

HVA ER SJANSEN?

I samarbeid med  **NR Norsk Regnesentral**

 **NORGE** [Bytt land](#)

 [Søk etter land](#)

 **LIVE** Sist beregnet: 57 d. siden

Neste beregning er ikke i gang ennå. Vi viser forrige snapshot til kjøringen starter.

 100 000 simuleringer 

VM: How to do something like this?



VM: Simulating scored goals

- Treat goals scored by each team as **independent Poisson RVs**.
- Suppose team A plays team B .
 - s_A and s_B strength parameters
 - n denote the expected number of goals scored by one team when two equally strong teams meet.

Then

$$X_A \sim \text{Poisson}(\lambda_A), \quad X_B \sim \text{Poisson}(\lambda_B),$$

with

$$\lambda_A = n \frac{s_A}{s_B}, \quad \lambda_B = n \frac{s_B}{s_A}.$$

VM: Match outcomes

Can calculate the probability of every possible scenario. For example,

$$P(X_A = i, X_B = j) = P(X_A = i)P(X_B = j) = \frac{\lambda_A^i}{i!} e^{-\lambda_A} \frac{\lambda_B^j}{j!} e^{-\lambda_B}, \quad i, j = 0, 1, \dots$$

The probability that A wins is therefore

$$P(A \text{ wins}) = \sum_{i>j} P(X_A = i, X_B = j),$$

and similarly one obtains probabilities for a draw or a B victory.

VM: Estimating unknown parameters

- Strengths $s_A, s_B, \dots$ and the baseline n have to be estimated.
- According to NR, they first used 3 experts assigned scores 1-100.
- As real tournament matches were played, those observations were incorporated into the estimates.
- **Maximum likelihood estimation** (like you learned in Statistics 1).

Solbakken om VM-beregninger: – Totalt ubrukelig

Ifølge datamodellene er Norge så godt som videre fra VM.
Tullprat, mener Ståle Solbakken.



[Ingvild Marie Svardal](#)
Journalist

+ 4 til

Publisert 22. juni kl. 01:57
Oppdatert 22. juni kl. 11:00

POENGJAKT: Ståle Solbakken tror ikke det holder for Norge med de tre poengene man har nå.

FOTO: FREDRIK VARFJELL / NTB

VM: Improvements

Issues?

- Independence assumption between the two teams' goal counts.
- Constant rate of scoring goals throughout the match.

Extensions?

- Separate attacking and defensive strengths.
- Home advantage.
- Dependence between the score counts of the two teams.
- Others?

Problem: P2 from Exercise 5

The number of goals scored by Vålerenga IF during a football match is Poisson distributed with an average of $\lambda_V = 1.2$ goals per match while the number of goals scored by Rosenborg BK is Poisson distributed with an average of $\lambda_R = 2$ goals per match. The number of goals scored by Vålerenga is independent of the number of goals scored by Rosenborg. Assume that a football match lasts for exactly 90 minutes (2×45 minutes) and that Vålerenga plays a match against Rosenborg:

- What is the distribution for the total number of goals scored in this match?
- What is the probability that there are no goals during the first half of the match?
- What is the probability that the final result is 2–2?
- What is the expected time until the first goal in this match?
- Assume that no goals are scored the first 15 minutes of the match. What is the probability that Vålerenga score at least one goal before the break at 45 minutes?

Problem: Exam

A certain scientific theory supposes that mistakes in cell division occur according to a Poisson process with rate 2.5 per year, and that an individual dies when 196 such mistakes have occurred. Assuming this theory, find

1.
 - The expected time until occurrence of the first mistake in cell division.
 - The expected lifetime of an individual.
 - The lifetime is uncertain. Quantify the uncertainty
2.
 - Calculate an approximation to the probability that an individual reaches an age of 90 years using the central limit theorem.
 - The probability that an individual reaches an age of 90 years can be calculated using Monte Carlo simulation. Describe an algorithm based on sampling from an exponential distribution.
3.
 - Determine a formula for an exact calculation of the probability that an individual reaches an age of 90 years using the Poisson probability distribution.
 - Determine a formula for an exact calculation of the probability that an individual reaches an age of 90 years using the gamma probability distribution.

Towards continuous-time Markov chains...

A Poisson process can be viewed as a Markov chain evolving in continuous time:

- The states are $\{0, 1, 2, \dots\}$.
- From state n , the process moves only to $n + 1$.
- The time spent in state n is $S_n \sim \text{Exp}(\lambda)$.
- By the memoryless property, the future depends only on the current state.

This is the basic idea behind **continuous-time Markov chains**:

- Random transitions occurring after exponential waiting times.
- More on that next week!